

8.

Ans: Bincrease in GPE of space probe = $GPE_{\text{final}} - GPE_{\text{initial}}$

$$= -\frac{GMm}{r_{\text{final}}} - \left(-\frac{GMm}{r_{\text{initial}}} \right) = (6.67 \times 10^{-11})(1.30 \times 10^{22})(5500) \left(\frac{1}{1.19 \times 10^7} - \frac{1}{(1.19 + 0.24) \times 10^7} \right)$$
$$= 6.73 \times 10^8 \text{ J}$$

9	Ans: D